FEM Beam Analysis Tool

Nodes

Number of nodes

3

Node 1 x

500.00

Node 1 y

0.00

Node 2 x

0.00

Node 2 y

0.00

Node 3 x

700.00

Node 3 y

-150.00

Properties (Material + Section)

Number of properties

1

Property 1

Property 1 name

Prop_ex1

Young's modulus E [Prop_ex1]

20000.00

Poisson's ratio v [Prop_ex1]

0.30

Section type [Prop_ex1]

general

Area [Prop_ex1]

60.00

Inertia [Prop_ex1]

500.00

Elements

Number of elements

2

Start node (Elem 1)

2

End node (Elem 1)

1

Element type (Elem 1)

Euler-Bernoulli 2-node

Property (Elem 1)

Prop_ex1

Start node (Elem 2)

1

End node (Elem 2)

3

Element type (Elem 2)

Euler-Bernoulli 2-node

Property (Elem 2)

Prop_ex1

Constraints

Number of constraints

6

Constraint 1 node	Constraint 1 direction	Constraint 1 value
2	x	0.00

Constraint 2 node	Constraint 2 direction	Constraint 2 value
2	y	0.00

Constraint 3 node	Constraint 3 direction	Constraint 3 value
2	rotation	0.00

Constraint 4 node	Constraint 4 direction	Constraint 4 value
3	x	0.00

Constraint 5 node	Constraint 5 direction	Constraint 5 value
3	y	0.00

Constraint 6 node	Constraint 6 direction	Constraint 6 value
3	rotation	0.00

Point Loads

Number of point loads

0

Distributed Loads

Number of distributed loads

1

Distributed Load 1 element	q_ini 1	q_fim 1	Direction 1
1	-0.40	-0.40	y

Moment Loads

Number of moment loads

1

Moment Load 1 node

1

Moment 1 magnitude

8000.00

Run Analysis

Nodal Displacements

	Node	x	y	u	v	theta
0	1	500	0	-0.091	-0.1993	0.0687
1	2	0	0	0	0	0
2	3	700	-150	0	0	0

Download Displacements CSV